UNIVERSITY OF CAPE COAST

COLLEGE OF AGRICULTURE AND NATURAL SCIENCES
SCHOOL OF PHYSICAL SCIENCES

DEPARTMENT OF MATHEMATICS

END OF FIRST SEMESTER EXAMINATION (2019/2020)

MAT 101: ALGEBRA AND TRIGONOMETRY

2 HOURS

Answer all questions in Section A and any two questions from Section B.

SECTION A

Answer all questions in this section.

1. Simplify $(P \cap Q') \cup (P \cap Q)$, where P and Q are subsets of the universal set μ.

   A. ϕ **B.** Q **C.** P **D.** μ

2. Solve $\log_2 \sqrt{2} = x$.

   A. $x = -\frac{1}{2}$ **B.** $x = \frac{1}{2}$

   C. $x = 1$ **D.** $x = 2$

3. All the following are characteristics of $\tan x$ except that

   A. its max. and min. values occurs at $\pm\infty$
   B. it is symmetrical about the origin
   C. it is periodic with period 2π
   D. it is asymptotic at $x = n\pi + \frac{\pi}{2}$

4. Find the difference d of an A.P for which $a_1 = 9$ and $S_5 = 15$.

   A. -5 **B.** -4 **C.** -3 **D.** 2

5. Form the polynomial equation whose zeros are $2 + \frac{1}{\sqrt{2}}, 2 - \frac{1}{\sqrt{2}}$.

   A. $x^2 - 3x - 1$ **B.** $2x^2 - 8x + 7$

   C. $2x^2 + x - 2$ **D.** $x^2 + 5x + 1$

6. The sixth term of the expansion $(x-2)^n$ is $-8064x^5$. Find the value of n.

   A. 5 **B.** 6 **C.** 10 **D.** 11

7. If in sets A, B, C, the set $B \cap C$ consists of 8 elements, set $A \cap B$ consists of 7 elements and set $C \cap A$ consists of 7 elements. Find the minimum element in the set $A \cup B \cup C$.

A. 8 **B.** 14 **C.** 22 **D.** 15

8. Solve the inequality $2\left|\frac{2}{3}x+1\right| \geq 4$.

A. $x \leq -\frac{9}{2}$ or $x \geq \frac{3}{2}$ **B.** $\frac{2}{3} < x \leq \frac{4}{9}$

C. $-\frac{9}{2} \leq x \leq \frac{3}{2}$ **D.** $x \geq \frac{3}{2}$

A function $f(x)$ is given by $f(x) = \frac{2x}{x^2+1}$.

Use it to answer Questions **9** to **12**.

9. Find the domain of $f(x)$.

A. $(-1, 1)$ **B.** $(-\infty, -1)$

C. $(-1, 0)$ **D.** $(-\infty, \infty)$

10. Find the range of $f(x)$.

A. $[-1, 1]$ **B.** $(-1, 1]$

C. $[-1, 1)$ **D.** $(-1, 1)$

11. Find the maximum turning point.

A. $(2, 1)$ **B.** $(-1, 1)$

C. $(1, 1)$ **D.** $(-1, -1)$

12. For what value of x is $f(x)$ undefined?

A. None **B.** $x = -1$

C. $x = 0$ **D.** $x = 1$

Use this information to answer Questions **13** and **14**.

The polynomial $P(x) = a_3x^3 + a_2x^2 + a_1x + a_0$ has zeros α, β and γ

13. Find a relation between the sum of the zeros and the coefficients of $P(x)$.

A. $\alpha+\beta+\gamma=\frac{a_1}{a_3}$ B. $\alpha+\beta+\gamma=-\frac{a_0}{a_3}$

C. $\alpha+\beta+\gamma=-\frac{a_2}{a_3}$ D. $\alpha+\beta+\gamma=\frac{a_2}{a_3}$

14. Which of the following relationships is correct?

A. $\alpha+\alpha\beta+\alpha\gamma=\frac{a_1}{a_3}$ B. $\alpha\beta+\beta\gamma+\alpha\gamma=\frac{a_1}{a_3}$

C. $\alpha\beta+\beta+\beta\gamma=\frac{a_1}{a_3}$ D. $\alpha\beta\gamma=\frac{a_0}{a_2}$

Use the equation $\mathbf{M}^2-3\mathbf{M}+5=0$ to answer Questions **15** and **16**.

15. Which of the following matrices satisfies the equation?

A. $\mathbf{M}=\begin{pmatrix}-1 & -3\\ -1 & 2\end{pmatrix}$ B. $\mathbf{M}=\begin{pmatrix}-1 & 3\\ 1 & -2\end{pmatrix}$

C. $\mathbf{M}=\begin{pmatrix}1 & -3\\ 1 & 2\end{pmatrix}$ D. $\mathbf{M}=\begin{pmatrix}1 & 3\\ 1 & 2\end{pmatrix}$

16. Find the inverse of the matrix $\mathbf{M}$.

A. $\begin{pmatrix}0.4 & 0.6\\ -0.2 & 0.2\end{pmatrix}$ B. $\begin{pmatrix}-0.4 & -0.6\\ -0.2 & 0.2\end{pmatrix}$

C. $\begin{pmatrix}-2 & 3\\ 1 & -1\end{pmatrix}$ D. $\begin{pmatrix}2 & 3\\ 1 & 1\end{pmatrix}$

17. Find the sum of the infinite series $\sum_{k=0}^{\infty}5\left(\frac{2}{3}\right)^k$.

A. 0 B. $\frac{10}{3}$ C. 10 D. 15

A function $f(x)$ is given by $f(x)=\frac{x^2-1}{x^2-4}$.

Use it to answer Questions **18** to **20**.

18. At what point does the curve cuts the x-axis?

A. $(1,0)$ and $(-1,0)$ **B.** $(1,0)$

C. $(-1,0)$ **D.** $\left(0,\frac{1}{4}\right)$

19. At what point does the curve cuts the y-axis?

A. $(1,0)$ and $(-1,0)$ **B.** $(1,0)$

C. $(-1,0)$ **D.** $\left(0,\frac{1}{4}\right)$

20. What observation can you draw from the function?

A. It is symmetrical about the origin
B. It has $y=0$ as the horizontal asymptote
C. It is symmetrical about the y-axis
D. It has $x=4$ as the vertical asymptote

21. Let X, Y and Z be sets such that $Y \subseteq X$. Which of the following is true?

A. $X \cup Z = Y \cup Z$ **B.** $Y' \cap Z' \subseteq X' \cap Z'$

C. $X \cap Z \subseteq Y \cap Z$ **D.** $X \cap (Y \cup Z) = Y \cup (X \cap Z)$

22. Find the Cartesian product of $(Z \times N) \cap (N \times Z)$, where N, Z are natural numbers and integers respectively .

A. $Z \times N$ **B.** $N \times Z$ **C.** $N \times N$ **D.** $Z \times Z$

Use the two equations $\log_y x - 4\log_x y = 0$ and $\log_{10} x + \log_{10} y = 1$ to answer Questions **23** and **24**.

23. Find the value of x.

A. 2 **B.** 5 **C.** $\sqrt{10}$ **D.** -10

24. Find the value of y.

A. 2 **B.** $\sqrt{5}$ **C.** $\sqrt{10}$ **D.** -10

25. In triangle ABC, right-angled at B, if $\tan A = \frac{1}{\sqrt{3}}$, find the value of $\sin A\cos C + \cos A\sin C$.

A. 0 **B.** 1 **C.** 2 **D.** 3

Use the matrix given as $\mathbf{Q} = \begin{bmatrix} -16 & -12 & 5 \\ 4 & 3 & -1 \\ -3 & -2 & 1 \end{bmatrix}$ to answer Question **26** to **29.**

26. Find the trace of the matrix $\mathbf{Q}$.

A. 20 B. 5 C. -12 D. 11

27. Find the sum of the minors of the diagonal elements of the matrix $\mathbf{Q}$.

A. 3 B. -4 C. 0 D. 1

28. Find the determinant of the matrix $\mathbf{Q}$.

A. 1 B. -19 C. -1 D. 19

29. Find the characteristic polynomial of the matrix $\mathbf{Q}$.

A. $x^3 - 5x^2 + 3x - 1$ B. $x^3 + 12x^2 + 1$

C. $x^3 + 11x^2 - 4x + 19$ D. $x^3 - 20x^2 + x - 19$

30. If θ is obtuse, and $\sin\theta = \frac{4}{5}$, find $\cos\theta$.

A. $-\frac{3}{4}$ **B.** $-\frac{3}{5}$ **C.** $\frac{3}{5}$ **D.** $\frac{3}{4}$

SECTION B

Answer two questions only from this section.
All questions carry equal marks.

1. (a) Sketch the graph of $f(x) = \sqrt{1-x^2}$.

Deduce the sketch of the graph of

$$h(x) = -2 + \sqrt{-(x+2)^2 + 1}$$

in the same plane.

(b) Show that if x is small enough to neglect its cube and higher powers,

$$\sqrt{\frac{1-x}{1+x}} = 1 - x + \frac{x^2}{2}.$$

(c) Find the truth set of $|x+2| \geq |x+1| + |x-3|$.

2. Consider the matrix

$$\mathbf{P} = \begin{pmatrix} 1 & -2 & 3 \\ -1 & 3 & 0 \\ 2 & -5 & 5 \end{pmatrix}.$$

(i) Find the characteristic polynomial of the matrix.

(ii) Express the inverse of **P** in the form

$\mathbf{P}^{-1} = \lambda\mathbf{P}^2 + \mu\mathbf{P} + \kappa\mathbf{I}$.

(iii) Hence, find the inverse matrix $\mathbf{P}^{-1}$.

(iv) Use your results to solve the system

$$x - 2y + 3z = 9$$

$$-x + 3y = -4$$

$$2x - 5y + 5z = 17$$

(b) Express the function $f(x) = \dfrac{x+1}{x^2 - x - 6}$ in its partial fractions.

3. (a) Sketch the parabola $y = x^2 - 4x + 5$.

(b) If the parabola is translated 3 units left and 2 units upwards, find,

(i) the equation of the translated curve.

(ii) the points of intersection of both curves.

(iii) the roots of the translated curve.

(c) Show that

$$(A \times B) \times C = A \times (B \times C)$$

if and only if at least one of the sets A, B or C is empty.

4. (a) A sequence $u_1, u_2, u_3, \ldots, u_n$ satisfies the relation

$$u_n = u_{n-1} + \left(\tfrac{1}{3}\right)^{n-1}, \ \forall n > 1 \text{ and } u_1 = 1.$$

(i) Write down the first four terms of the sequence.

(ii) Obtain an expression for S_n and hence, find S_{10}.

(b) State and prove the Zero Multiplication Theorem.

End of Paper